**3.**

$$f(x) = 3 - \frac{x-2}{x+1} + \frac{5x+26}{2x^2-3x-5} \qquad x > 4$$

(a) Show that

$$f(x) = \frac{ax+b}{cx+d} \qquad x > 4$$

where $a$, $b$, $c$ and $d$ are integers to be found.

**(4)**

(b) Hence find $f^{-1}(x)$

**(2)**

(c) Find the domain of $f^{-1}$

**(2)**